

Final Project Report

Course: SWAPD 402 – Mobile Application Development

Project Name: Bite Bright

Technology: Flutter & Dart

By: yassmin Raafat

Pakinam khaled

Sama Reda

Sandy mohamed

1. Introduction

App Purpose

Bite Bright is a mobile application designed to help users maintain a healthy and balanced lifestyle by tracking their daily meals and encouraging mindful eating habits based on the **80/20 principle**, which promotes balance rather than restriction.

Target Users

The application targets:

- Individuals interested in maintaining a healthy diet
- Users who want a simple and organized way to track meals
- People seeking awareness and motivation regarding their eating habits

Overall Functionality

The application allows users to:

- Log daily meals
 - Track nourishing versus flexible meals
 - Monitor progress and balance over time
 - Authenticate securely using Firebase Authentication
-

2. Project Phases Overview

Phase 1 – App Concept and Basic UI

In this phase, the foundational structure of the application was developed:

- Flutter project setup and configuration
 - Main navigation implemented using a Bottom Navigation Bar
 - Responsive UI designed using standard Flutter widgets
 - Consistent color schemes, typography, and spacing applied
-

Phase 2 – Core Functionality and Data Handling

This phase focused on application logic and data management:

- Meal logging functionality implemented
 - State management handled using the Provider pattern
 - Local data storage implemented using Shared Preferences
 - API integration added to fetch daily wellness tips
 - Local notifications implemented to enhance user engagement
-

Phase 3 – Finalization, Optimization, and Deployment

Animations and Transitions

To enhance the user experience:

- UI animations were added for smoother interaction
- Screen transitions were implemented for better navigation flow
- Animated widgets were used, especially on the Home screen

Performance Optimization

Several optimization techniques were applied:

- `IndexedStack` was used to prevent unnecessary widget rebuilds
- Navigation performance was improved
- Bugs were identified and fixed using Flutter debugging tools

Firebase Authentication

Firebase services were integrated into the application:

- Email and password authentication implemented
- Login and registration screens created
- User session managed through authentication state

App Icon and Splash Screen

Final UI enhancements included:

- Custom application icon added
- Native splash screens configured
- Final UI polished with consistent colors, layouts, and typography

Deployment

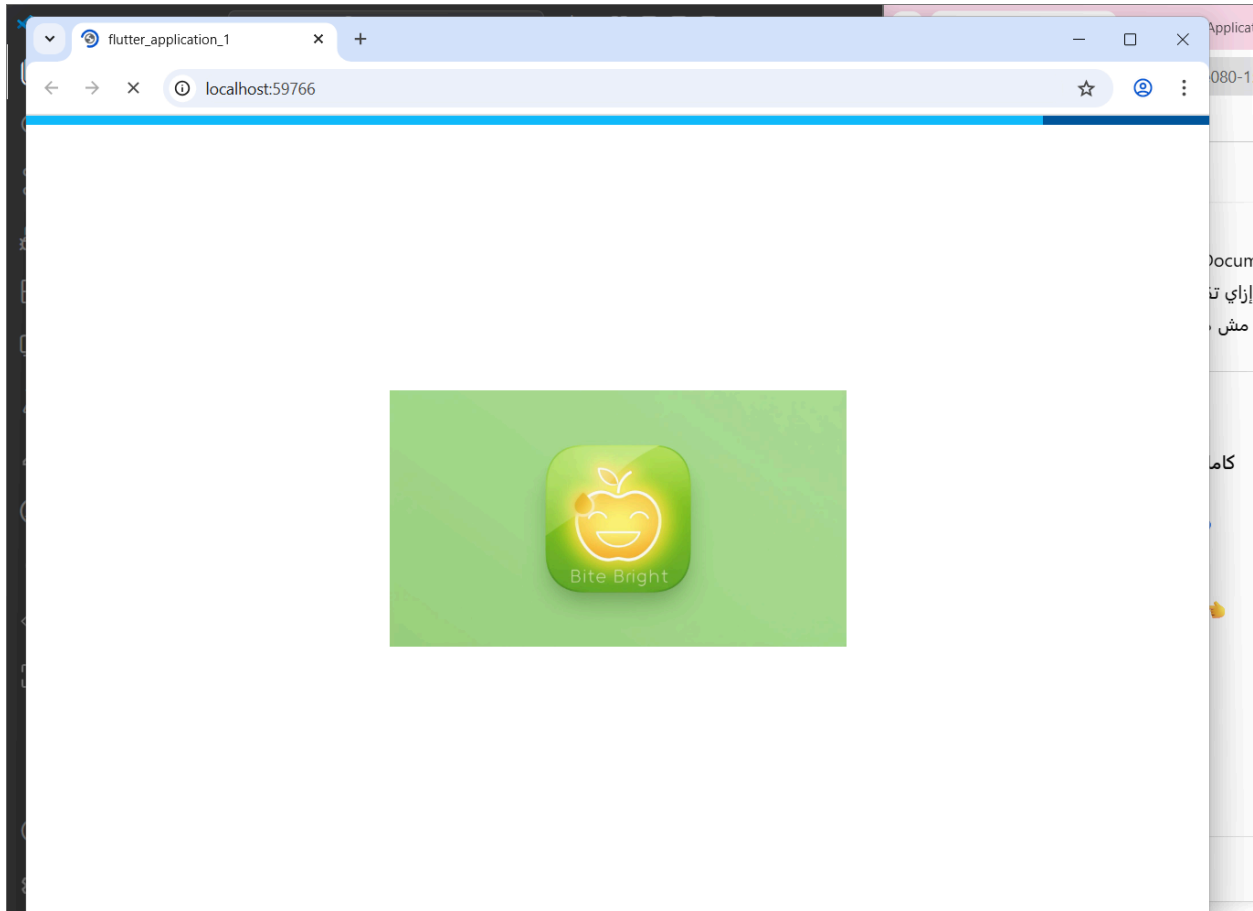
- A release APK was generated successfully
 - The application is ready for installation and demonstration
-

3. Tools and Technologies Used

- **Flutter** – Cross-platform mobile application framework
- **Dart** – Programming language
- **Provider** – State management solution
- **Firebase Authentication** – Secure user authentication
- **Shared Preferences** – Local data storage
- **Flutter Local Notifications** – User notifications
- **REST API** – Daily wellness tips
- **GitHub** – Version control and collaboration

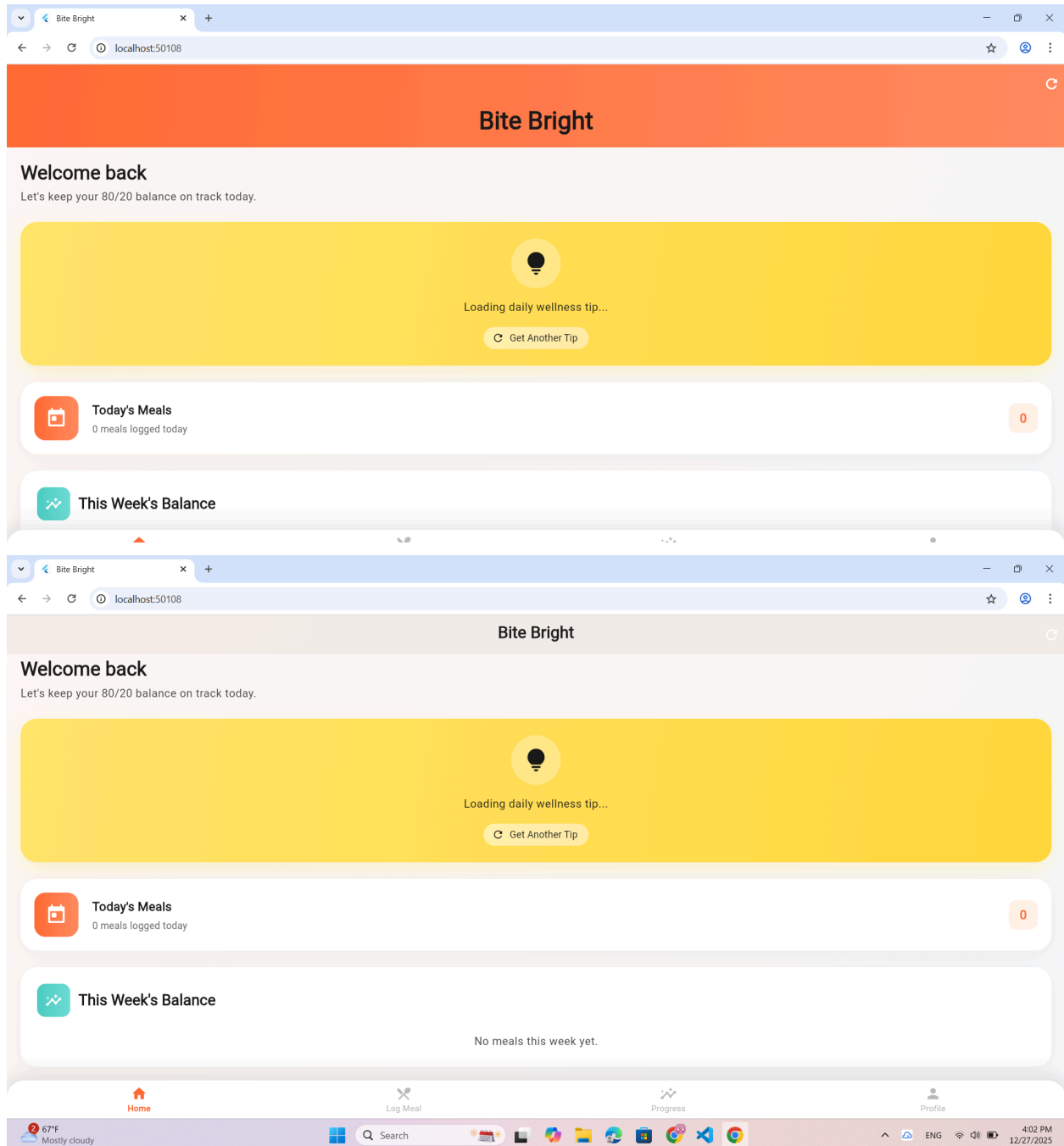
Splash Screen

This screen represents the initial loading screen of the application. It displays the Bite Bright logo and branding while the app initializes, providing a smooth and professional first impression for users.



Home Screen

The Home screen provides an overview of the application and encourages users to follow the 80/20 healthy eating principle. It acts as the main entry point and navigation hub, offering a clean and welcoming interface.



Log Meal Screen

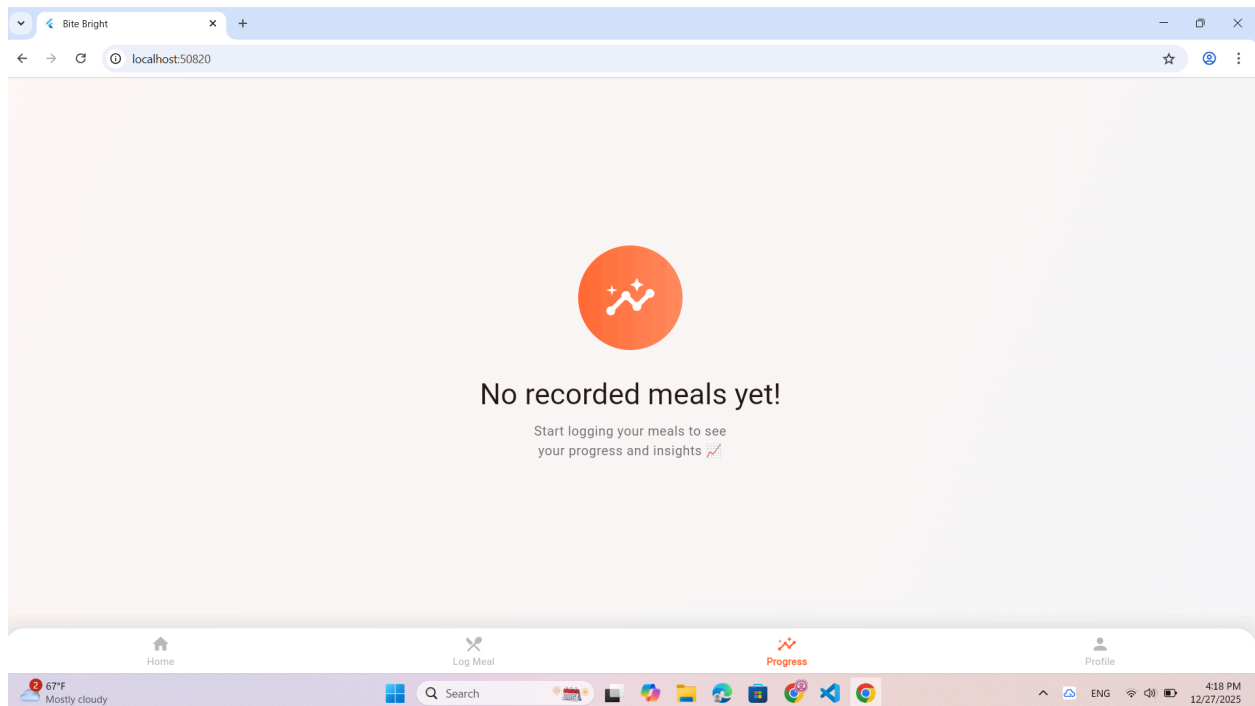
Figure X: Log Meal Screen

This screen allows users to log their daily meals. Users can select the meal type (Nourishing or Flexible), enter meal details, and optionally record how they felt after eating. This supports mindful eating and accurate meal tracking.

The screenshot displays the 'Log Meal' screen in a web browser. The browser's address bar shows 'localhost:50820'. The page has an orange header with the title 'Log Meal' and a close button. Below the header, there's a section for 'Meal Type (80/20 Rule)' with two buttons: 'Nourishing' (80% Healthy) and 'Flexible' (20% Enjoyment). The 'Nourishing' button is highlighted in teal. Below this is a 'Meal Description' section with a text input field labeled 'What did you eat?'. Underneath is an optional section 'How did you feel? (Optional)' with a text input field labeled 'E.g., Energized, Satisfied, Happy...'. A large orange button with a plus sign and the text 'Add Meal' is positioned below the input fields. At the bottom, there's a 'Your Meals' section with a circular icon and the text 'Your Meals' next to a counter showing '0'. The bottom of the screen shows a navigation bar with icons for Home, Log Meal (active), Progress, and Profile. The system tray at the very bottom shows the date and time as 4:17 PM on 12/27/2025.

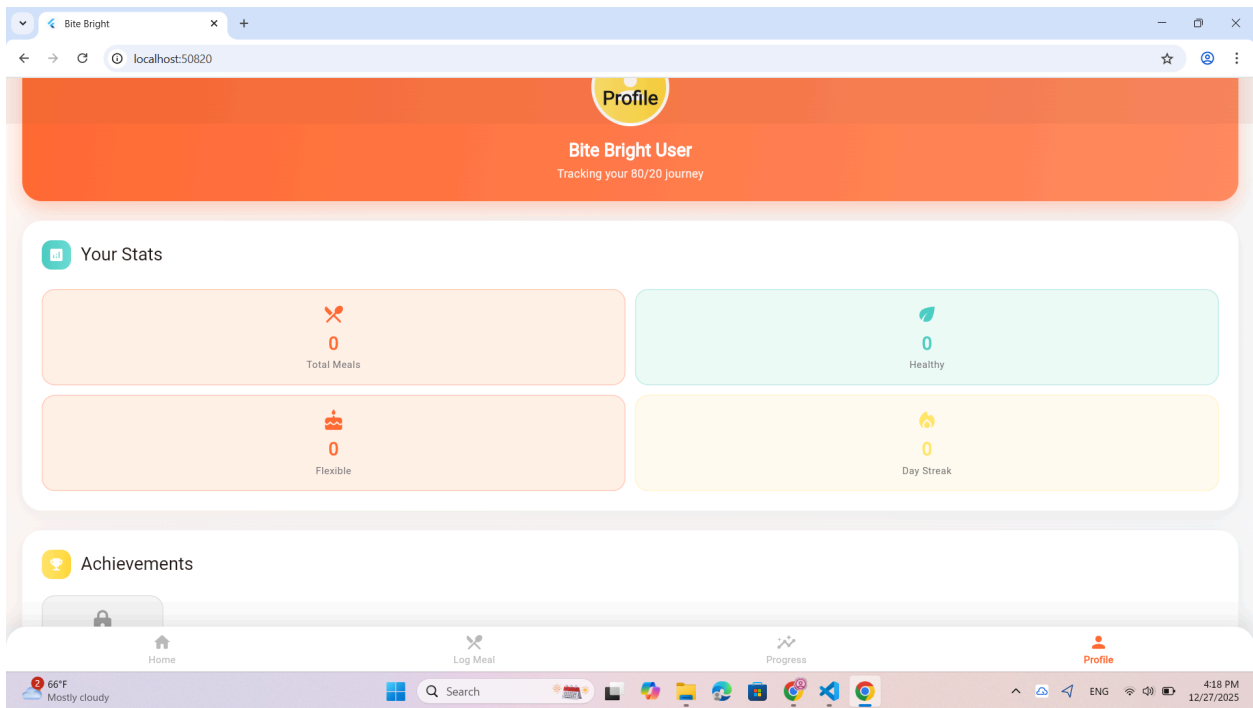
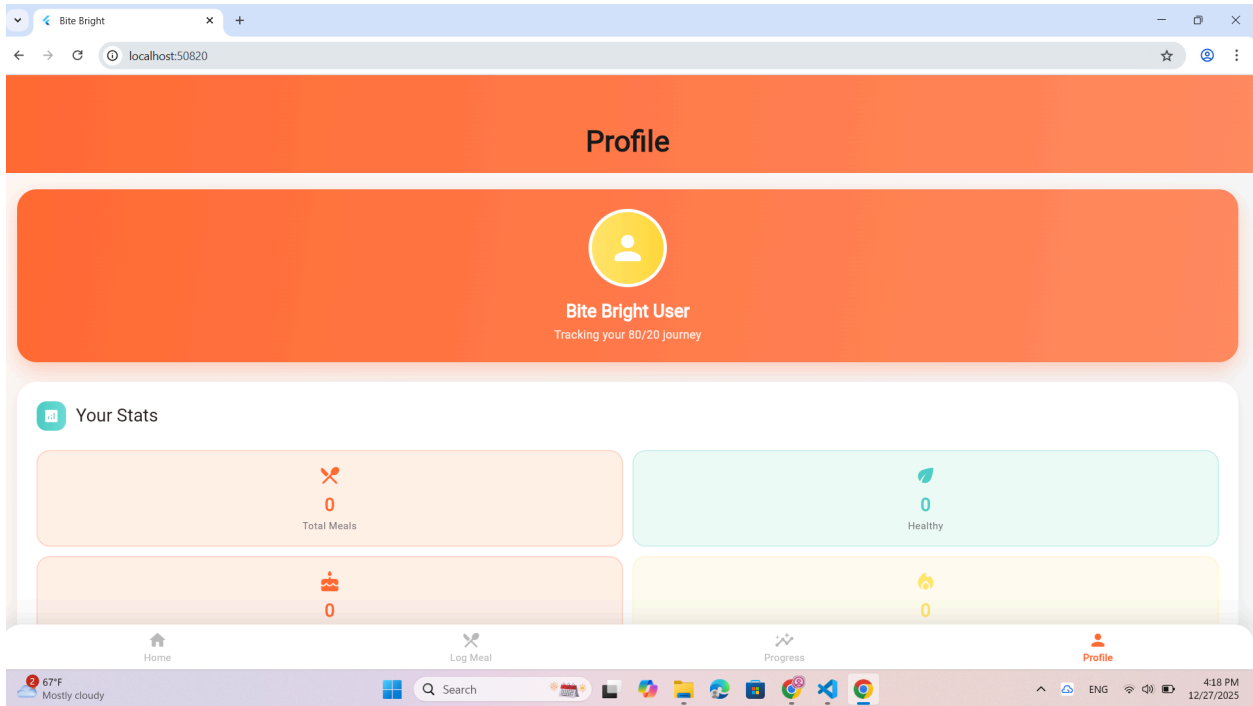
Progress Screen

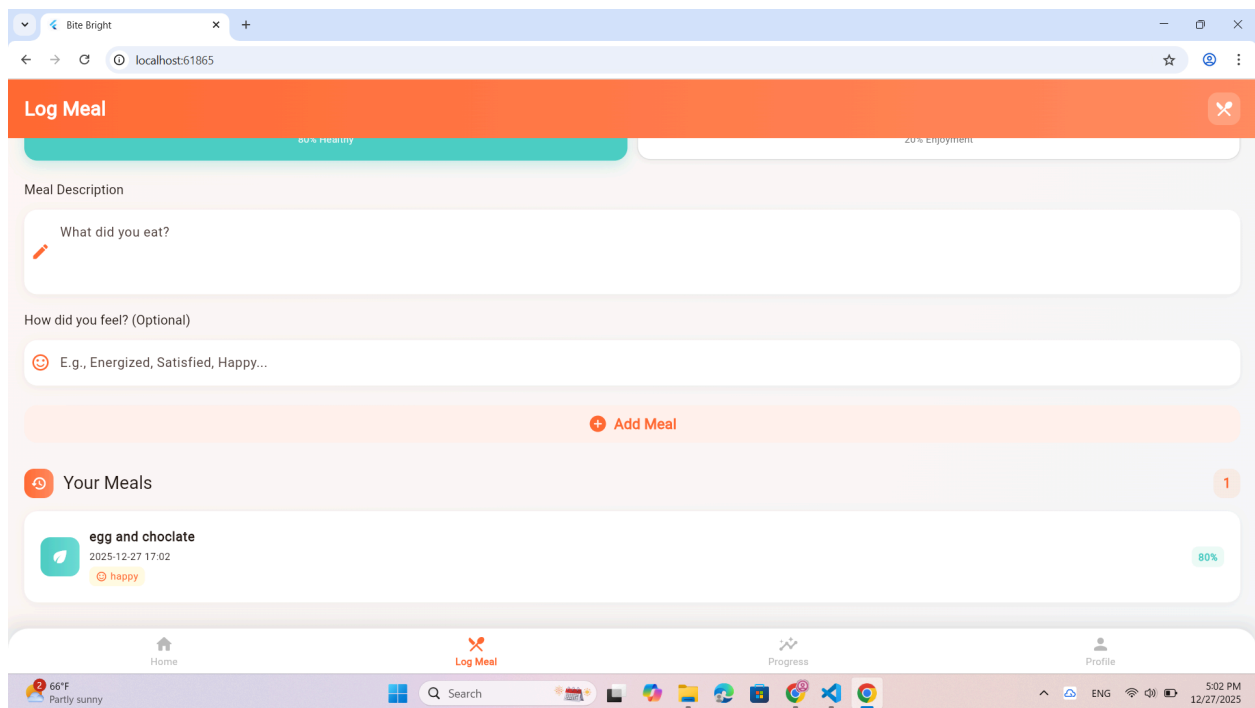
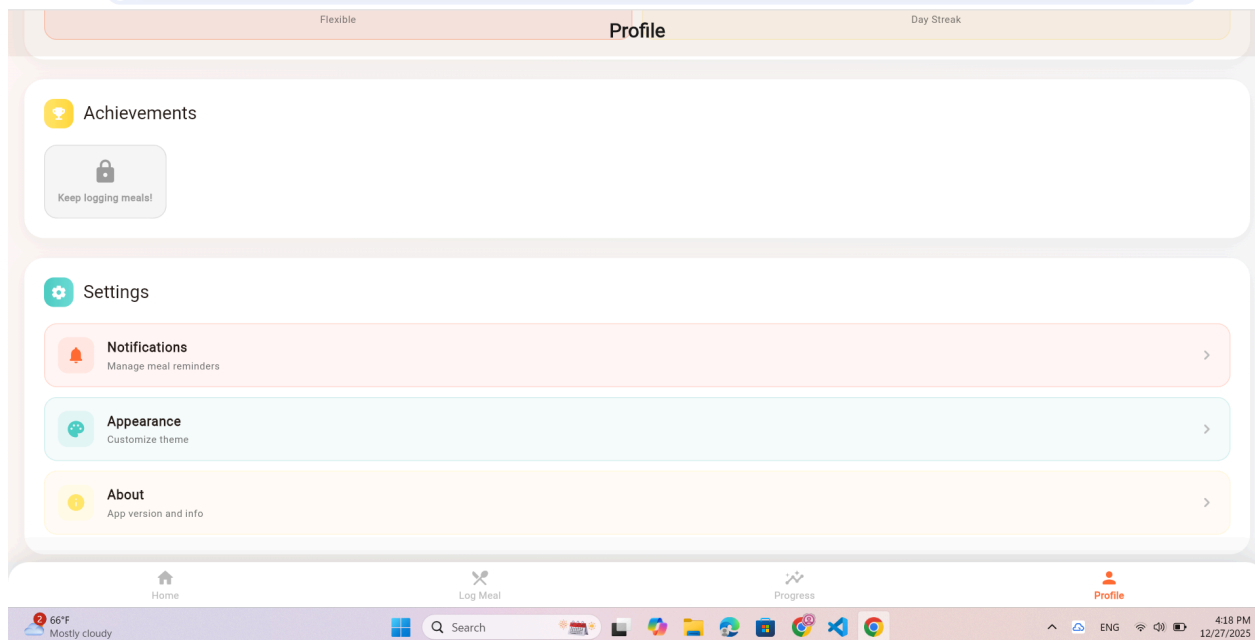
The Progress screen visualizes the user's meal tracking journey. It provides insights into logged meals and encourages consistency by displaying progress-related information in a clear and user-friendly layout.

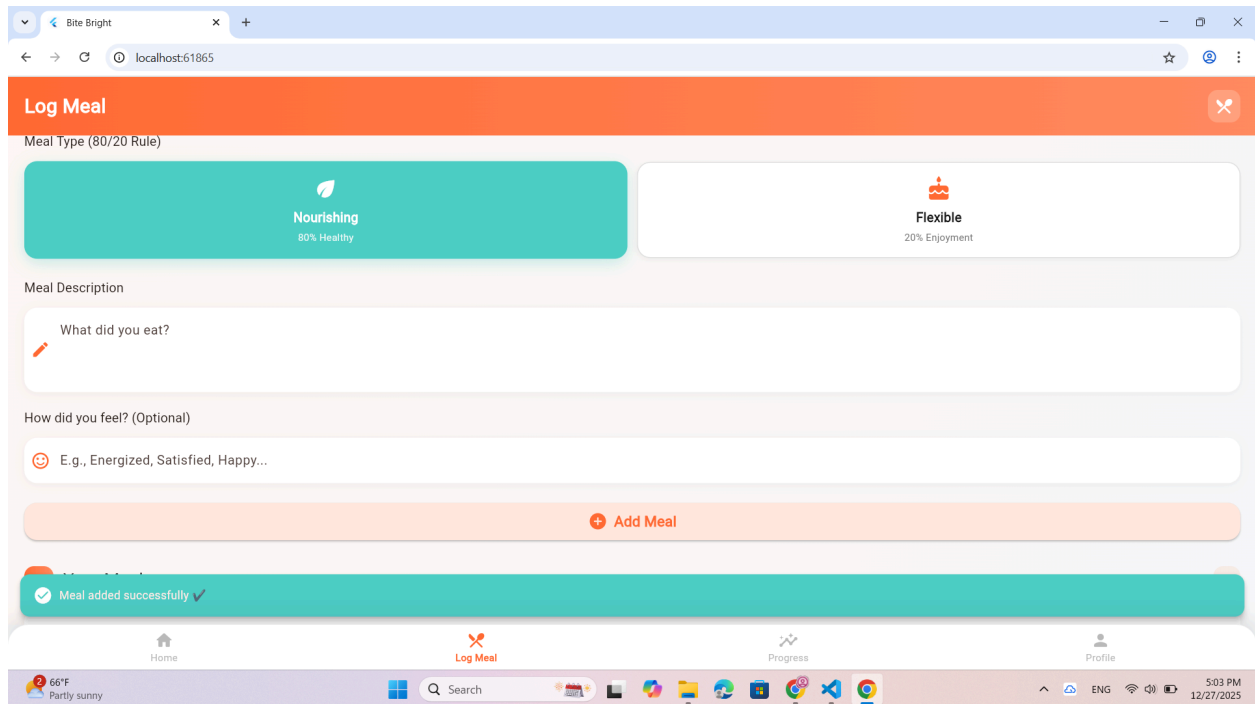


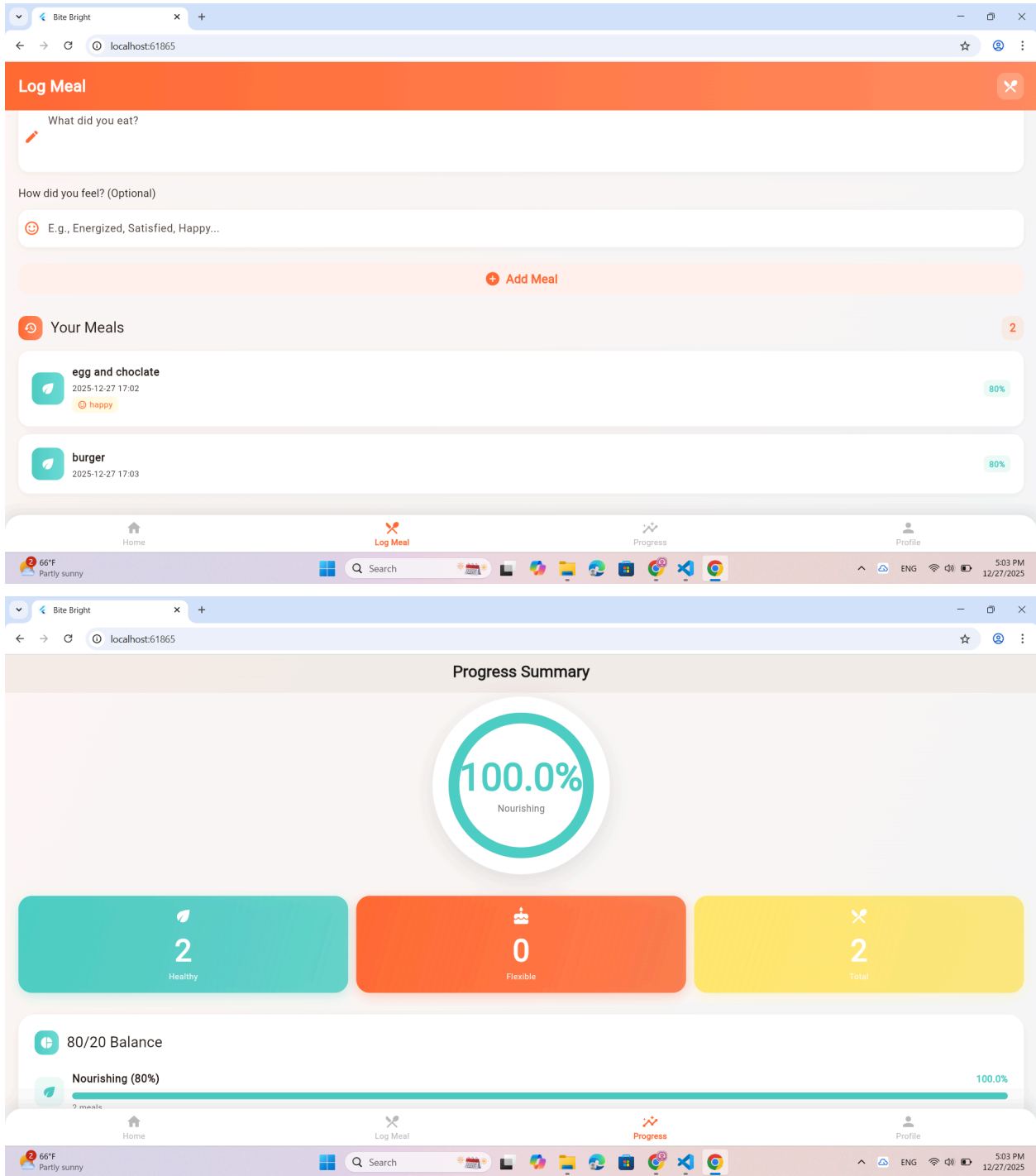
Profile Screen

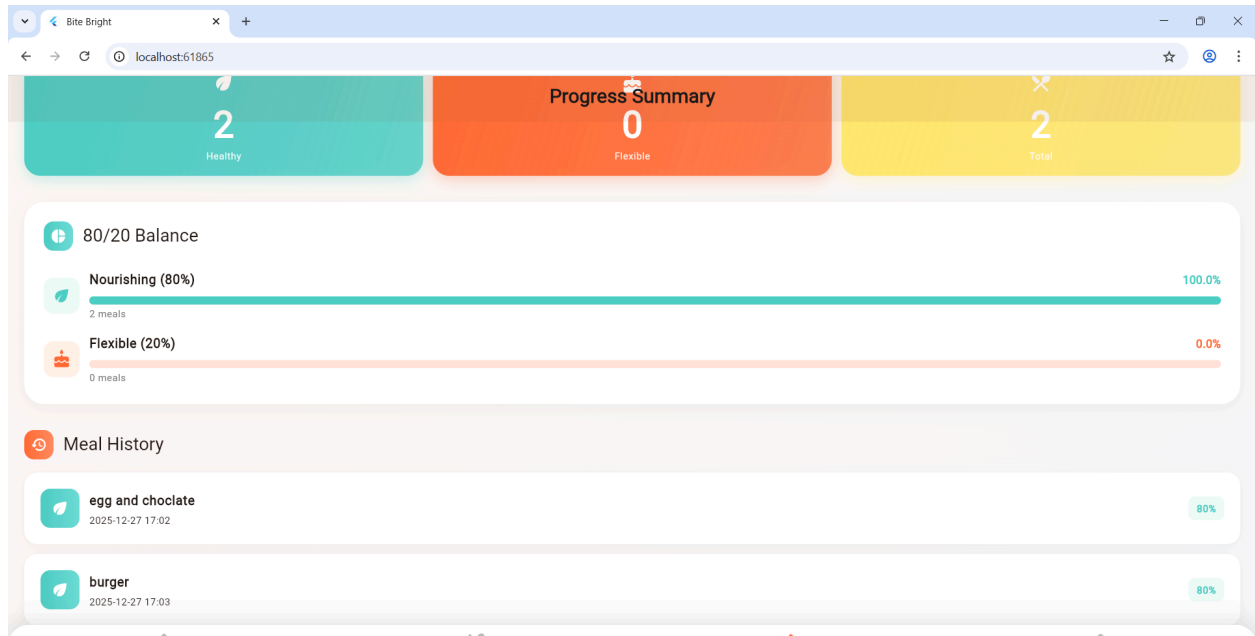
The Profile screen displays user information and summarizes personal statistics such as total meals, healthy meals, flexible meals, and day streaks. It helps users monitor their overall performance and engagement with the app.











5. Conclusion

Bite Bright successfully demonstrates the development of a complete mobile application using Flutter and Dart. The project integrates user-friendly UI design, efficient state management, Firebase authentication, performance optimization, and deployment. The application fulfills all course requirements and provides a practical solution for promoting healthy and mindful eating habits.